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ISOLATED BATTERY DETECTION

Abstract

In some examples, a circuit includes an amplifier and a comparator. The amplifier has an output terminal, first and second input terminals, and an enable terminal, the second input terminal of the amplifier coupled to the output terminal of the amplifier. The comparator has an output terminal and first and second input terminals, the first input terminal of the comparator coupled to the first input terminal of the amplifier, and the output terminal of the comparator coupled to the enable terminal of the amplifier.

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Background/Summary

BACKGROUND

[0001] Various electronic devices include a battery for powering the device untethered from a wired power adapter or power source. However, the electronic devices may also be powered by the

wired power adapter concurrently, or in the absence of the battery.

SUMMARY

[0002] In some examples, a circuit includes an amplifier and a comparator. The amplifier has an output terminal, first and second input terminals, and an enable terminal, the second input terminal of the amplifier coupled to the output terminal of the amplifier. The comparator has an output terminal and first and second input terminals, the first input terminal of the comparator coupled to the first input terminal of the amplifier, and the output terminal of the comparator coupled to the enable terminal of the amplifier.

[0003] In some examples, a circuit includes a power converter, a battery detection circuit, and a controller. The power converter has an input terminal, an output terminal, and a control terminal, the input terminal coupled to an input voltage terminal of the circuit, and the output terminal coupled to a system voltage terminal of the circuit. The battery detection circuit has first and second input terminals and first and second output terminals, the first input terminal of the battery detection circuit coupled to a battery voltage terminal of the circuit, and the first output terminal of the battery detection circuit coupled to a battery driver terminal of the circuit. The controller has an input terminal and first and second output terminals, the input terminal of the controller coupled to the second output terminal of the battery detection circuit, the first output terminal of the controller coupled to the second input terminal of the battery detection circuit, and the second output terminal of the controller coupled to the control terminal of the power converter.

[0004] In some examples, a system includes a battery charger circuit and a switch. The battery charger circuit includes a power converter, a battery detection circuit, and a controller. The power converter has an input terminal, an output terminal, and a control terminal, the input terminal coupled to an input voltage terminal of the battery charger circuit and the output terminal coupled to a system voltage terminal of the battery charger circuit. The battery detection circuit has first and second input terminals and first and second output terminals, the first input terminal of the battery detection circuit coupled to a battery voltage terminal of the battery charger circuit, and the first output terminal of the battery detection circuit coupled to a battery driver terminal of the battery charger circuit. The controller has an input terminal and first and second output terminal, the input terminal of the controller coupled to the second output terminal of the battery detection circuit, the output terminal of the controller coupled to the second input terminal of the battery detection circuit, and the second output terminal of the controller coupled to the control terminal of the power converter. The switch has a control terminal and having first and second terminals, the control terminal of the switch coupled to the battery driver terminal, the first terminal of the switch coupled to the system voltage terminal, and the second terminal of the switch coupled to the battery voltage terminal.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIG. 1 is a block diagram of a system, in accordance with various examples.

[0006] FIG. 2 is a block diagram of a battery charger circuit, in accordance with various examples.

[0007] FIG. 3 is a block diagram of a battery detection circuit, in accordance with various examples.

[0008] FIG. 4 is a diagram of signal waveforms, in accordance with various examples.

[0009] FIG. 5 is a flow diagram of a method, in accordance with various examples.

DETAILED DESCRIPTION

[0010] As described above, various electronic devices include a battery for powering the device untethered from a wired power adapter or power source. In the absence of the battery, the electronic devices may continue operation, powered by the wired power adapter. Similarly, both the wired

power adapter and the battery may be coupled to the electronic devices at a same time, with components of the electronic devices receiving power from the wired power adapter, the battery, or a combination of both. It may therefore be useful to determine whether the battery is or is not coupled to the electronic device at a particular point in time, for example, to maintain a stable and regulated system voltage based on power received from the wired power adapter. Various approaches may exist for performing such a detection. However, these approaches may face various challenges. For example, some approaches may modify the system voltage to determine whether the battery is coupled to the electronic device, thereby destabilizing the system voltage for other components of the electronic device. In other examples, a detection of whether the battery is coupled to the electronic device may rely on the implementation of additional pins on a integrated circuit die and/or package, or the implementation of optional external components coupled to the integrated circuit. These approaches may increase a size, and thereby cost, of the integrated circuit, rendering the integrated circuit unsuitable for certain application environments.

[0011] Examples of this disclosure provide for isolated battery detection. In some examples, the battery detection may be performed within an integrated circuit without the addition of other components external to the integrated circuit beyond those implemented in the absence of the isolated battery detection. The isolated battery detection may be performed in a manner that does not perturb the system voltage (e.g., a voltage being provided by the integrated circuit to other components of the electronic device). In some examples, the isolated battery detection is performed by modulating a voltage provided by the integrated circuit to a battery terminal based on a reference signal, where the modulation does not alter the system voltage. In some examples, the reference signal is a time-varied signal, such as signal having a periodic sinusoidal waveform, triangular waveform, trapezoidal waveform, square waveform, or any other suitable waveform shape. Responsive to detecting a change in voltage at the battery terminal less than or approximately equal to the reference signal (e.g., the voltage at the battery terminal is changing approximately proportionally to the reference signal), the integrated circuit determines that the battery is not present. Responsive to detecting a change in voltage at the battery terminal greater than the reference signal, the integrated circuit determines that the battery is present.

[0012] FIG. 1 is a block diagram of a system **100**, in accordance with various examples. The system **100** may be generally representative of any device which receives a direct current (DC) input voltage (VIN), provides a DC system voltage (VSYS), and couples to a battery at which a battery voltage (VBAT) is provided. In an example, the system **100** includes a power source **102**, a battery charger circuit **104**, a battery **106**, and a component **108**. The power source **102** may be any power source capable of providing VIN as a DC signal, such as an alternating current (AC) to DC adapter or converter, a solar or other renewable power source, a battery, a capacitor, or the like. In some examples, the power source **102** is included in a same device (e.g., in a same housing) as a remainder of the system **100**, while in other examples the power source **102** is external to a remainder of the components of the system **100**. The battery charger circuit **104** may include circuitry suitable for receiving VIN, providing VSYS having a value greater than, less than, or approximately equal to VIN, controlling charging of the battery **106**, and performing isolated battery detection, as described herein. The battery **106** may be any detachable or removable battery of any suitable chemistry or architecture, the scope of which is not limited herein, such that the battery **106** may be deemed “replaceable” or “user-replaceable.” In some examples, the battery **106** may be implemented as a battery module or other device that includes additional components or circuitry beyond simply a power storage structure. In some examples, the battery **106** may be a capacitor, such as a super-capacitor, or another form of energy storage device known by a term other than “battery” but which charges, stores, and discharges energy. The component **108** may be any component which receives VSYS and operates based on VSYS. For example, the component **108** may be a sensor, a camera, a speaker, a display, a microphone, a DC motor, a processing circuit such as a processor, microprocessor, controller, microcontroller, or the like. In some examples, the

component **108** may be functional for a particular range (e.g., safe operating range) of values of VSYS, where VSYS having a value outside of that safe operating range may cause the component **108** to cease expected functionality, such as resetting, restarting, shutting down, or operating in an unknown, and potentially uncontrolled, state.

[0013] In an example, the battery charger circuit **104** performs isolated battery detection. For example, the battery **106** may be coupled by a switch **110** to a node at which VSYS is provided. The switch **110** may be controlled by the battery charger circuit **104**. While shown in FIG. **1** as being separate from the battery charger circuit **104**, in some examples the switch **110** may be a component of, and included with a package or housing of, the battery charger circuit **104**. In some examples, the switch **110** is a transistor, such as a n-channel metal oxide semiconductor field effect transistor (NMOS). In other examples, the switch **110** is a bi-polar junction transistor. In yet other examples, the switch **110** is a mechanical or electromechanical relay. Generally, the switch **110** may be any controllable device, controllable by the battery charger circuit **104** to electrically couple, or decouple, the battery **106** from the node at which the battery charger circuit **104** provides VSYS.

[0014] The battery charger circuit **104** may generate, or receive, a reference signal (REF) (as shown in FIG. **2**). The reference signal may be time-varied, such as a periodic signal of any suitable waveform shape. In some examples, the battery charger circuit **104** generates the reference signal internally, such as via components included in the package of the battery charger circuit **104**. In other examples, the battery charger circuit **104** receives the reference signal from any suitable source, the scope of which is not limited herein. Based on the reference signal, the battery charger circuit **104** modulates the providing of VSYS to the battery **106**. For example, the battery charger circuit **104** may control conductivity of the switch **110** based on the reference signal. In an example of the system **100** in which the switch **110** is a NMOS, the reference signal may cause the switch **110** to operate in a low-dropout (LDO) mode in which electrical characteristics of the switch **110** approximate a resistor coupled between the node at which VSYS is provided at the battery **106**. Thus, by controlling the conductivity of the switch **110**, the battery charger circuit **104** controls an effective series resistance of the switch **110**. By controlling the effective series resistance of the switch **110**, the battery charger circuit **104** modulates a voltage provided to the battery **106** to have a waveform shape and value approximately equal to the reference signal.

[0015] The battery charger circuit **104** includes a terminal (e.g., pin, node, etc.) at which the battery **106** couples. This terminal may have a parasitic capacitance that is inherent to the terminal and occurs irrespective of whether a discrete capacitive component is coupled to the terminal. The parasitic capacitance may, however be comparatively small in value. In other examples, a discrete component (not shown) having a non-zero amount of capacitance may be coupled to the terminal of the battery charger circuit **104** at which the battery **106** couples. The capacitance presented at the terminal of the battery charger circuit **104** at which the battery **106** couples, in the absence of the battery **106**, may be referred to herein as a parasitic capacitance irrespective of whether the capacitance results from including a discrete component coupled to the terminal, the capacitance being inherently occurring, or a combination of both. In some examples, the parasitic capacitance has a value in a range of approximately 1 microfarad (uF) to approximately 100 uF. In some examples, the parasitic capacitance has a value of approximately 10 uF. As a result, in the absence of the battery **106** (e.g., responsive to the battery **106** being decoupled from the battery charger circuit **104**), a voltage provided at the terminal may change approximately proportionally to the reference signal and responsive to a change in value of the reference signal. Thus, in the absence of the battery **106**, the voltage provided at the terminal may approximately equal the reference signal. However, in the presence of the battery **106**, the voltage provided at the terminal may not change approximately proportionally to the reference signal. For example, the battery **106** may present at the terminal as a large capacitor, which may therefore take an increased amount of time to change in value in comparison to the parasitic capacitance.

[0016] Because of the difference in behavior of the voltage provided at the terminal in the absence,

or presence, of the battery **106**, the voltage provided at the terminal may be used to determine whether the battery **106** is present. For example, the battery charger circuit **104** may compare the reference signal to the voltage provided at the terminal. Responsive to the reference signal being greater than, or equal to, the voltage provided at the terminal, the battery charger circuit **104** may determine that the battery **106** is decoupled from the battery charger circuit **104**. Conversely, responsive to the reference signal being less than the voltage provided at the terminal, the battery charger circuit **104** may determine that the battery **106** is coupled to the battery charger circuit **104**. However, irrespective of whether the battery **106** is or is not coupled to the battery charger circuit **104**, VSYS may be unperturbed by the battery detection operations. Thus, battery detection of the battery **106** by the battery charger circuit **104** may be isolated from VSYS and the component **108**. [0017] FIG. 2 is a block diagram of the battery charger circuit **104**, in accordance with various examples. Although the battery charger circuit **104** of FIG. 2 is described in the context of the system **100** of FIG. 1, in various examples the battery charger circuit **104** may be suitable for implementation in other system, components, devices, or the like. In an example, the battery charger circuit **104** includes a power converter **202** and a battery detection circuit **204**. In some examples, the power converter **202** is a buck power converter, a boost power converter, a buck-boost power converter, a low-dropout regulator, a switch mode power converter, or any other circuit or component(s) suitable for receiving VIN and providing VSYS based on VIN. In an example, the battery detection circuit **204** includes a VIN terminal **206**, a VSYS terminal **208**, a battery driver terminal **210**, and a VBAT terminal **212**.

[0018] The power converter **202** may have an input coupled to the VIN terminal **206** and an output coupled to the VSYS terminal **208**. In some examples, the power converter **202** also couples via other terminals (not shown) to other components (not shown) which may be external to the battery charger circuit **104**, such as an inductor. The battery detection circuit **204** has an input coupled to the VBAT terminal **212** and an output coupled to the battery driver terminal **210**. In some examples, the switch **110** has a control terminal (e.g., a gate in an example of a MOSFET) coupled to the battery driver terminal **210**, a first terminal (e.g., a drain in an example of a MOSFET) coupled to the VSYS terminal **208**, and a second terminal (e.g., a source in an example of a MOSFET) coupled to the VBAT terminal **212**. In some examples, a capacitor **211** is coupled between the VBAT terminal **212** and a ground node at which a ground voltage potential is provided. The capacitor may have a capacitance of about 10 uF, as described above. In some examples, the capacitor **211** is included in the battery charger circuit **104**, while in other examples the capacitor **211** is external to the battery charger circuit **104**. In some examples, a resistor **213** is coupled between second terminal of the switch **110** and the VBAT terminal **212**. The resistor **213** may be a sense resistor, such as enabling detection of an amount of current flowing through the switch **110**. The resistor **213** may also limit an amount of current flowing through the switch **110** into the battery **106** in examples in which the battery **106** is present.

[0019] In some examples, the power converter **202** and the battery detection circuit **204** are controlled by a controller **214**. The controller **214** may be included in the battery charger circuit **104**, as shown in FIG. 2, or may be external to and coupled to the battery charger circuit **104**. The controller **214** may control the power converter **202** by providing control signals to control conductivity of switching components (not shown), such as transistors, of the power converter **202**. Generally, the controller **214** may control the power converter **202** and the power converter **202** may operate to provide VSYS based to VIN according to any suitable process(es), the scope of which is not limited herein.

[0020] The controller **214** may control the battery detection circuit **204** by providing a reference signal for controlling the switch **110**, and may receive a battery detection signal (BATOK) from the battery detection circuit **204**. In some examples, the controller **214** provides the reference signal to the battery detection circuit **204** directly. In other examples, the controller **214** controls a reference circuit (not shown) to provide the reference signal to the battery detection circuit **204**. The battery

detection circuit **204** of FIG. 2 may operate in a manner suitable for performing the isolated battery detection described above with respect to the battery charger circuit **104** of FIG. 1, and such description is not repeated herein with respect to FIG. 2.

[0021] FIG. 3 is a block diagram of the battery detection circuit **204**, in accordance with various examples. Although the battery detection circuit **204** of FIG. 3 is described in the context of the system **100** of FIG. 1 and battery charger circuit **104** of FIG. 2, in various examples the battery detection circuit **204** may be suitable for implementation in other system, components, devices, or the like. For example, the battery detection circuit **204** may be implemented to detect a plugin event of a conductor or connector having a capacitance substantially larger than a parasitic capacitance provided at a terminal or which the battery detection circuit **204** and the connector couple. In some examples, the battery detection circuit **204** as shown in FIG. 3 may be implemented in a separate electrical component package or housing than a remainder of the battery charger circuit **104** and may couple to the battery charger circuit **104**.

[0022] In an example, the battery detection circuit **204** includes an amplifier **302** and a comparator **304**. In some examples, the amplifier **302** may be a gated amplifier, such that the amplifier **302** provides an output signal responsive to receipt of an asserted enable signal, and blocks or does not provide the output signal responsive to receive of the enable signal having a deasserted value (or, more generally, in the absence of the asserted enable signal). As used herein, the enable signal may be asserted when the enable signal has a logic-level high (e.g., digital logic “1”) value.

[0023] The amplifier **302** has a first input terminal (e.g., a positive or non-inverting input terminal) configured to receive the reference signal. For example, the amplifier **302** may couple to the controller **214** via the first input terminal of the amplifier **302**. The amplifier **302** also has a second input terminal (e.g., a negative or inverting input terminal) and an output terminal. The output terminal of the amplifier **302** is coupled to the second input terminal of the amplifier **302**. The output terminal of the amplifier **302** may also couple to the battery driver terminal **210**. The comparator **304** has a first input terminal (e.g., a positive or non-inverting input terminal) coupled to the first input terminal of the amplifier **302** (e.g., to receive the reference signal), a second input terminal coupled to the VBAT terminal **212**, and an output terminal. In an example, the comparator **304** provides BATOK at the output terminal of the comparator **304**. In some examples, the output terminal of the comparator **304** is coupled to an enable terminal of the amplifier **302** such that BATOK is the enable signal for the amplifier **302**. In some examples, the output terminal of the comparator **304** is coupled to the controller **214**, such as to provide BATOK to the comparator **304**. In some examples, BATOK may be an active-low signal. For example, BATOK having a logic-level high (e.g., digital logic “1”) value may indicate that the battery **106** is not coupled to the battery charger circuit **104**, and BATOK having a logic-level low (e.g., digital logic “0”) value may indicate that the battery **106** is coupled to the battery charger circuit **104**. The battery detection circuit **204** of FIG. 3 may operate in a manner suitable for performing the isolated battery detection described above with respect to the battery charger circuit **104** of FIG. 1 and battery detection circuit **204** of FIG. 2, and such description is not repeated herein with respect to FIG. 3.

[0024] In some examples, battery detection operations of the battery detection circuit **204** may be performed under control of the controller **214**. For example, the controller **214** may implement a digital state machine, or other executable, programmable, or logic function, to determine a value for REF. In an example, responsive to BATOK transitioning from a logic-level high value to a logic-level low value, the controller **214** may cause REF to transition to having a value of approximately 0 V. In this way, battery detection operations of the battery detection circuit **204** may be disabled by the controller **214**. The controller **214** may subsequently resume providing VREF having a modulated value, as described above herein, such as responsive to receipt of another signal by the controller **214**, the scope of which is not limited herein. In some examples, the signal may indicate that the capacitor **211** has been charged and/or discharged greater than a threshold number of times within a threshold amount of time. For example, responsive to the controller **214**

determining that the capacitor **211** has been charged and/or discharged greater than the threshold number of times within the threshold amount of time, the controller **214** may determine that the battery **106** is not present. Responsive thereto, the controller **214** may re-enable the battery detection operations of the battery detection circuit **204**.

[0025] FIG. **4** is a diagram **400** of signal waveforms, in accordance with various examples. In an example, the diagram **400** shows ideal waveforms such that timing variations and transient responses that may occur in one or more signals represented in the diagram **400** are not present in the waveforms shown in FIG. **4**. The diagram **400** includes VSYS, VBAT, REF, and BATOK.

[0026] As described above and shown by the diagram **400**, responsive to the battery **106** being decoupled from the battery charger circuit **104**, VBAT has a value approximately equal to REF and BATOK has a logic-level high value. As REF changes in value under modulation by the battery detection circuit **204**, VBAT changes proportionally and responsively thereto in the absence of the battery **106**. Responsive to battery **106** being coupled to the battery charger circuit **104**, VBAT has a value greater than REF and BATOK transitions to a logic-level low value. The battery charger circuit **104** may subsequently begin charging the battery **106**, further increasing the value of VBAT with respect to REF. Although not shown in FIG. **4**, responsive to a subsequent decoupling of the battery **106** from the battery charger circuit **104**, VBAT may decrease in value to become less than REF, and BATOK may transitions to again have a logic-level high value.

[0027] FIG. **5** is a flow diagram of a method **500**, in accordance with various examples. In some examples, the method **500** is implemented in a system, such as the system **100**, to determine the presence or absence of a battery, such as the battery **106**. The method **500** may be implemented in whole or in part by a battery charger circuit, such as the battery charger circuit **104**. For example, the method **500** may be implemented in whole or in part by a battery detection circuit, such as the battery detection circuit **204**, of, or coupled to, the battery charger circuit.

[0028] At operation **502**, a reference signal is provided. In some examples, the reference signal is a time-varied signal, such as signal having a periodic sinusoidal waveform, triangular waveform, trapezoidal waveform, square waveform, or any other suitable waveform shape.

[0029] At operation **504**, a switch is driven based on the reference signal. In some examples, the switch is driven by the reference signal to modulate a voltage provided at a battery voltage terminal. The switch may be, for example, a MOSFET transistor operated in an LDO mode to approximate electrical characteristics of a resistor.

[0030] At operation **506**, the reference signal is compared to the voltage provided at the battery voltage terminal to determine a comparison result. In examples in which the battery is not present, the reference signal may have a value greater than or equal to the voltage provided at the battery voltage terminal. In such an example, the comparison result may have a logic-level high value. In examples in which the battery is present, the reference signal may have a value less than the voltage provided at the battery voltage terminal. In such an example, the comparison result may have a logic-level low value.

[0031] In this description, the term “couple” may cover connections, communications, or signal paths that enable a functional relationship consistent with this description. For example, if device A generates a signal to control device B to perform an action: (a) in a first example, device A is coupled to device B by direct connection; or (b) in a second example, device A is coupled to device B through intervening component C if intervening component C does not alter the functional relationship between device A and device B, such that device B is controlled by device A via the control signal generated by device A.

[0032] A device that is “configured to” perform a task or function may be configured (e.g., programmed and/or hardwired) at a time of manufacturing by a manufacturer to perform the function and/or may be configurable (or reconfigurable) by a user after manufacturing to perform the function and/or other additional or alternative functions. The configuring may be through firmware and/or software programming of the device, through a construction and/or layout of

hardware components and interconnections of the device, or a combination thereof.

[0033] A circuit or device that is described herein as including certain components may instead be coupled to those components to form the described circuitry or device. For example, a structure described as including one or more semiconductor elements (such as transistors), one or more passive elements (such as resistors, capacitors, and/or inductors), and/or one or more sources (such as voltage and/or current sources) may instead include only the semiconductor elements within a single physical device (e.g., a semiconductor die and/or integrated circuit (IC) package) and may be coupled to at least some of the passive elements and/or the sources to form the described structure either at a time of manufacture or after a time of manufacture, for example, by an end-user and/or a third-party.

[0034] While certain components may be described herein as being of a particular process technology, these components may be exchanged for components of other process technologies. Circuits described herein are reconfigurable to include the replaced components to provide functionality at least partially similar to functionality available prior to the component replacement. Components shown as resistors, unless otherwise stated, are generally representative of any one or more elements coupled in series and/or parallel to provide an amount of impedance represented by the shown resistor. For example, a resistor or capacitor shown and described herein as a single component may instead be multiple resistors or capacitors, respectively, coupled in parallel between the same nodes. For example, a resistor or capacitor shown and described herein as a single component may instead be multiple resistors or capacitors, respectively, coupled in series between the same two nodes as the single resistor or capacitor.

[0035] Uses of the phrase “ground voltage potential” in the foregoing description include a chassis ground, an Earth ground, a floating ground, a virtual ground, a digital ground, a common ground, and/or any other form of ground connection applicable to, or suitable for, the teachings of this description. In this description, unless otherwise stated, “about,” “approximately” or “substantially” preceding a parameter means being within ± 10 percent of that parameter. Modifications are possible in the described examples, and other examples are possible within the scope of the claims.

[0036] As used herein, the terms “terminal,” “node,” “interconnection,” “pin,” and “lead” are used interchangeably. Unless specifically stated to the contrary, these terms are generally used to mean an interconnection between or a terminus of a device element, a circuit element, an integrated circuit, a device, or a semiconductor component. Furthermore, a voltage rail or more simply a “rail,” may also be referred to as a voltage terminal and may generally mean a common node or set of coupled nodes in a circuit at the same potential.

Claims

1. A circuit, comprising: an amplifier having an output terminal, first and second input terminals, and an enable terminal, the second input terminal of the amplifier coupled to the output terminal of the amplifier; and a comparator having an output terminal and first and second input terminals, the first input terminal of the comparator coupled to the first input terminal of the amplifier, and the output terminal of the comparator coupled to the enable terminal of the amplifier.
2. The circuit of claim 1, wherein the second input terminal of the comparator is coupled to a battery voltage terminal of a battery charger circuit.
3. The circuit of claim 1, wherein the output terminal of the amplifier is coupled to a battery driver terminal of a battery charger circuit.
4. The circuit of claim 1, wherein the output terminal of the comparator is coupled to a controller.
5. The circuit of claim 4, wherein the controller is configured to provide a reference signal at the first input terminal of the amplifier.
6. The circuit of claim 1, wherein the amplifier is configured to receive a reference signal at the first input terminal of the amplifier.

7. The circuit of claim 6, wherein the reference signal is a time-varied signal.

8. The circuit of claim 6, wherein the amplifier is configured to provide a signal at the output terminal of the amplifier configured to modulate a voltage provided at the second input terminal of the comparator.

9. A circuit, comprising: a power converter having an input terminal, an output terminal, and a control terminal, the input terminal coupled to an input voltage terminal of the circuit, and the output terminal coupled to a system voltage terminal of the circuit; a battery detection circuit having first and second input terminals and first and second output terminals, the first input terminal of the battery detection circuit coupled to a battery voltage terminal of the circuit, and the first output terminal of the battery detection circuit coupled to a battery driver terminal of the circuit; and a controller having an input terminal and first and second output terminals, the input terminal of the controller coupled to the second output terminal of the battery detection circuit, the first output terminal of the controller coupled to the second input terminal of the battery detection circuit, and the second output terminal of the controller coupled to the control terminal of the power converter.

10. The circuit of claim 9, wherein the battery detection circuit comprises: an amplifier having an output terminal, first and second input terminals, and an enable terminal, the output terminal of the amplifier coupled to the battery driver terminal and to the second input terminal of the amplifier, and the first input terminal of the amplifier coupled to the first output terminal of the controller; and a comparator having an output terminal and first and second input terminals, the first input terminal of the comparator coupled to the first input terminal of the amplifier, and the output terminal of the comparator coupled to the enable terminal of the amplifier and to the input terminal of the controller.

11. The circuit of claim 10, wherein the power converter is configured to provide a system voltage signal at the system voltage terminal responsive to receipt of an input voltage provided at the input voltage terminal.

12. The circuit of claim 11, wherein the battery detection circuit is configured to modulate a voltage provided at the battery driver terminal of the circuit based on a reference signal received at the first input terminal of the amplifier to induce a change in voltage at the battery voltage terminal.

13. The circuit of claim 12, wherein the battery detection circuit is configured to compare the reference signal to a voltage provided at the battery voltage terminal to determine a comparison result indicating presence or absence of a battery coupled to the battery voltage terminal, wherein the comparison result indicating that the reference signal has a value less than the voltage provided at the battery voltage terminal indicates that the battery is coupled to the battery voltage terminal.

14. A system, comprising: a battery charger circuit, comprising: a power converter having an input terminal, an output terminal, and a control terminal, the input terminal coupled to an input voltage terminal of the battery charger circuit and the output terminal coupled to a system voltage terminal of the battery charger circuit; a battery detection circuit having first and second input terminals and first and second output terminals, the first input terminal of the battery detection circuit coupled to a battery voltage terminal of the battery charger circuit, and the first output terminal of the battery detection circuit coupled to a battery driver terminal of the battery charger circuit; and a controller having an input terminal and first and second output terminal, the input terminal of the controller coupled to the second output terminal of the battery detection circuit, the output terminal of the controller coupled to the second input terminal of the battery detection circuit, and the second output terminal of the controller coupled to the control terminal of the power converter; and a switch having a control terminal and having first and second terminals, the control terminal of the switch coupled to the battery driver terminal, the first terminal of the switch coupled to the system voltage terminal, and the second terminal of the switch coupled to the battery voltage terminal.

15. The system of claim 14, wherein the battery detection circuit comprises: an amplifier having an output terminal, first and second input terminals, and an enable terminal, the output terminal of the

amplifier coupled to the control terminal of the switch and to the second input terminal of the amplifier, and the first input terminal of the amplifier coupled to the first output terminal of the controller; and a comparator having an output terminal and first and second input terminals, the first input terminal of the comparator coupled to the first input terminal of the amplifier, the output terminal of the comparator coupled to the enable terminal of the amplifier and to the input terminal of the controller, and the second input terminal of the comparator coupled to the battery voltage terminal.

16. The system of claim 15, wherein the battery detection circuit is configured to: receive a reference signal; and modulate a signal provided at the control terminal of the switch to control an equivalent series resistance of the switch, wherein modulating the signal provided at the control terminal of the switch causes a signal having a same waveform shape as the reference signal to be provided at the second terminal of the switch.

17. The system of claim 16, wherein the reference signal is a time-varied signal.

18. The system of claim 16, wherein the battery detection circuit is configured to: detect a battery voltage signal at the second input terminal of the comparator; and compare the reference signal to the battery voltage signal to determine a comparison result, wherein the comparison result has a logic-level low value responsive to the battery voltage signal being greater in value than the reference signal.

19. The system of claim 18, wherein the battery detection circuit is configured to determine that a battery is coupled to the battery voltage terminal responsive to the comparison result having the logic-level low value.

20. The system of claim 16, wherein the battery detection circuit is configured to control the switch to provide the signal having the same waveform shape as the reference signal at the second terminal of the switch without affecting a voltage value of a system voltage provided at the system voltage terminal.
